A Quantity Theory of Money Approach to Cryptocurrency Valuation: Store of Value, Medium of Exchange, and the Economic Moat Signal

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Abstract

We apply the Quantity Theory of Money (QTM) to the cross-sectional valuation of cryptocurrencies and protocol tokens. Treating each digital asset as the currency of a self-contained economy, we decompose market capitalization into a Medium of Exchange (MoE) component, priced by the free-circulating velocity of supply, and a Store of Value (SoV) residual. The key innovation is λ—the fraction of total supply that is locked or staked—which partitions the circulating supply into economically active and inert components. We derive a "free velocity" (V_{free}) applicable only to unlocked supply, yielding a cleaner estimate of the MoE component and, by subtraction, the SoV premium. For native blockchain coins, high λ reflects a convenience yield and seigniorage signal grounded in monetary search theory. For governance and utility tokens, high λ carries a distinct meaning: it is an economic moat signal, capturing token holder conviction in platform growth and foreclosing the exit option. We construct two summary metrics—the NVT/Growth ratio and the SoV/MoE ratio—and arrange assets in a quadrant framework that identifies undervalued, overvalued, and moat-endowed assets. Using a panel of [N] major cryptocurrencies and tokens over [sample period], we find that quadrant positioning significantly predicts future returns, with "Star" assets (high SoV/MoE, low NVT/G) generating superior risk-adjusted returns over 12–24 month horizons.

1 Introduction

What determines the value of a cryptocurrency? The dominant approach in the literature has been to adapt discounted cash flow (DCF) models, searching for a dividend-like flow—transaction fees, staking rewards, governance rights—that can be capitalized into a present value (Cong, Li, and Wang, 2021; Liu and Tsyvinski, 2021). This approach, however, mischaracterizes the fundamental nature of digital assets. Cryptocurrencies and protocol tokens are not equity claims on future earnings. They are currencies. Each coin or token is the monetary instrument of a self-contained digital economy, and its value is determined by the same forces that determine the value of any currency: the scale of economic activity it enables and the efficiency with which it circulates. The correct framework, we argue, is the Quantity Theory of Money.

The Fisher equation, $MV = PQ$, relates the stock of money (M) and its velocity (V) to the nominal value of economic output (PQ). Applied to digital assets, a cryptocurrency's market capitalization proxies its money supply, and the economic output it supports—on-chain transaction volume for native coins, protocol throughput for tokens—is its PQ. Under this mapping, a coin's market capitalization satisfies MC $\approx PQ/V$, making value a direct function of economic activity and circulation efficiency. This is not merely an analogy. Cong, Li, and Wang (2021) develop a formal dynamic model in which token prices aggregate heterogeneous users' transactional demand, with equilibrium prices governed by platform adoption rather than discounted cash flows. Sockin and Xiong (2020) show that token price appreciation equals the total cost of capital and platform participation minus the convenience yield accruing to marginal users—a formulation structurally identical to the QTM framework we develop.

The QTM framework, however, requires a critical extension. Not all of a cryptocurrency's circulating supply is economically active. A substantial and growing fraction is locked in staking contracts, pledged as governance collateral, or held in long-term custody with near-zero velocity. These holdings are economically inert: they contribute to the money supply but perform no transactional function. Conflating active and inactive supply distorts velocity estimates and, consequently, misprices the economic content of market capitalization. We address this by introducing λ, defined as the proportion of the total token supply that is locked or staked:

$$\lambda = \frac{M_{\text{locked}}}{M}.$$

The free-circulating supply is then $(1-\lambda)M$, and we define the *free velocity* V_{free} as the velocity of only this active portion. This yields the MoE component of market capitalization as

$$\text{MC}_{\text{MoE}} = \frac{PQ}{V_{\text{free}}},$$

and, by residual, the Store of Value component as

$$\text{MC}_{\text{SoV}} = \text{MC} - \text{MC}_{\text{MoE}}.$$

MC_{SoV} captures everything the market assigns to the asset beyond its current transactional utility: long-term holding premia, network security premia, option value over future platform growth, and speculative demand. The ratio $\text{MC}_{\text{SoV}}/\text{MC}_{\text{MoE}}$ is our primary measure of an asset's strategic balance between speculative conviction and current utility.

Why does λ matter beyond its mechanical role in adjusting velocity? For native blockchain coins—Ethereum, Solana, and their peers—high λ is grounded in three complementary theoretical traditions. First, *monetary search theory* (Kiyotaki and Wright, 1993) establishes that money emerges as a coordination equilibrium: an asset becomes valuable because rational agents expect others to hold and accept it. High λ is the blockchain-observable, publicly verifiable signal of this coordination. Unlike traditional monetary systems where holding preferences are latent, on-chain data makes coordination commitments transparent and measurable. Second, the *convenience yield* literature (Sockin and Xiong, 2020) establishes that holders of dominant monetary assets earn a non-pecuniary return—a premium for access to network liquidity, security participation, and future transactional capability—

just as holders of U.S. Treasuries earn a safety and liquidity premium beyond their yield (Krishnamurthy and Vissing-Jorgensen, 2012). High λ signals that a large share of rational agents expect the network's convenience yield to grow, and they are paying today for future network membership. Third, the *seigniorage* channel (Cong and He, 2024) completes the picture: stakers in proof-of-stake networks capture new token issuance as block rewards, making the staking decision a revealed preference over the network's positive net present value. These three channels are complementary: coordination signals why high λ is self-reinforcing, convenience yield explains the holding motive, and seigniorage grounds the rational optimization underlying the staking decision.

Governance and utility tokens present a related but distinct case. A governance token such as UNI is not the settlement currency of Ethereum's transaction economy; it is the currency of the Uniswap protocol economy. Its PQ is not on-chain gas consumption but the total throughput of the Uniswap protocol—decentralized exchange volume, liquidity deployed, fees generated. Whether protocol revenue currently accrues to token holders is immaterial to this characterization, just as it is immaterial whether U.S. GDP "flows to" dollar holders. The dollar has value because the U.S. economy is large and growing. UNI has value because the Uniswap economy is large and growing. This framing aligns with Cong, Li, and Wang (2021), who price tokens on the present value of future platform utility and adoption rather than current cash flows. For governance tokens, however, λ carries a different signal than for coins. Staking a native coin to validate a network provides security and earns seigniorage—a function with clear monetary parallels. Locking a governance token into a DAO contract and casting votes is an act of conviction about the protocol's economic trajectory. Drawing on Hirschman (1970), holders who exercise *voice* (governance participation) rather than *exit* (selling) reveal a preference for the protocol's future that passive buyers do not. Edmans and Manso (2011) formalize this in a corporate setting: blockholders who engage rather than exit create governance value precisely because engagement is costly and therefore credible. Applied to tokens, active governance participation—staking plus voting—is a costly signal

(Spence, 1973) of platform conviction, and high participation rates are an economic moat indicator. Recent empirical evidence confirms this: DAO Governance (2024) document that proposal passage increases DAO token returns by 4.7%, and that a one-standard-deviation increase in voting participation amplifies this effect by 2.2%.

We operationalize these ideas in two summary metrics and an investment framework. The *NVT/Growth ratio* (NVT/G) normalizes the Network Value to Transactions ratio by the network's growth rate, yielding a PEG-ratio analog for digital assets: assets with low NVT/G are cheap relative to their adoption trajectory. The *SoV/MoE ratio* captures the asset's strategic balance between speculative conviction and current utility. Plotting assets on a 2×2 matrix across these dimensions yields four quadrants with distinct investment implications: "Star/Economic Moat" assets (high SoV/MoE, low NVT/G) combine long-term conviction with undervalued transactional utility; "GARP" assets (low SoV/MoE, low NVT/G) offer utility-driven value at reasonable prices; "Avoid" assets (high SoV/MoE, high NVT/G) combine speculation with expensive utility; and "Mature/Overpriced" assets (low SoV/MoE, high NVT/G) have high current utility priced at a premium. We find that quadrant positioning strongly predicts future cross-sectional returns, with Star assets generating [X]% higher annual returns than Avoid assets over [horizon], after controlling for size, momentum, and market beta.

Related Literature. Our paper contributes to several strands of the literature. The first is the theory and empirics of cryptocurrency valuation. Cong, Li, and Wang (2021) develop a dynamic model of token adoption and valuation grounded in network externalities; Sockin and Xiong (2020) model tokens as platform membership with convenience yield pricing; Liu and Tsyvinski (2021) document systematic risk factors in cryptocurrency returns. We complement these by providing an empirically tractable decomposition of market capitalization grounded in QTM, and by constructing testable investment signals from on-chain observables. The second strand is the monetary economics of digital assets. Kiyotaki and Wright

(1993) and subsequent search-theoretic models of money inform our treatment of λ as a coordination signal; Cong and He (2024) analyze the tokenomics of proof-of-stake staking; the convenience yield literature (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016) motivates the monetary premium embedded in $\mathrm{MC}_{\mathrm{SoV}}$; Jiang, Krishnamurthy, and Lustig (2024) and the reserve currency literature provide the analogy between the dollar's exorbitant privilege and the dominant-network premium in crypto. The third strand is DAO governance and token holder behavior. Edmans and Manso (2011) formalize voice versus exit in a blockholder setting; DAO Governance (2024) and DAO Review (2025) document empirical patterns in DAO voting and their asset pricing consequences. Our paper is the first, to our knowledge, to connect governance participation rates to a QTM-based valuation decomposition and to use them as an investment signal.

The remainder of the paper is organized as follows. Section **??** develops the formal theoretical framework, derives the SoV/MoE decomposition, and presents formal propositions distinguishing coins from governance tokens. Section **??** states the testable hypotheses. Section **??** describes the data, sample construction, and variable measurement. Section **??** presents the main empirical results. Section **??** reports robustness tests. Section **??** concludes.

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